

ELECTENG 209

Electronics System Design

Signal Conditioning

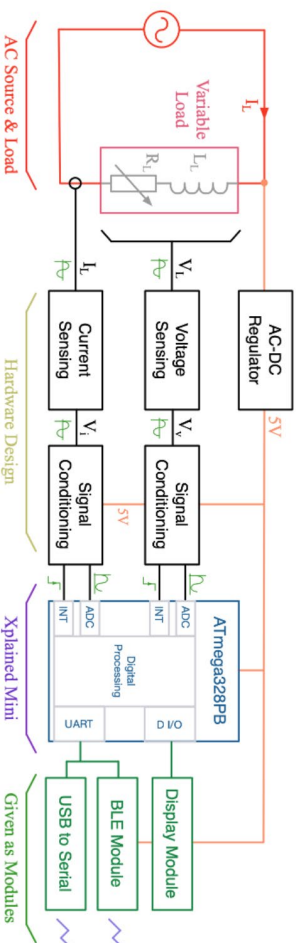
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Learning Objectives

- Review of operational amplifier basics
- What are inverting and non-inverting amplifiers
- How are difference amplifiers created and why are they useful?
- Understanding practical signal conditioning circuits using single sided supplies
- What practical considerations are needed with voltage and current measurement circuits?
- How do we create RC filters and what are the design approaches in the frequency domain?
- Understanding the use and design of comparators for interrupt circuits
- Design guidelines for creating a regulated supply suited to your analogue and digital electronics.

The Energy Monitor Hardware design

- Requires Voltage Sensing, Current Sensing as discussed earlier.
- Signal conditioning
 - Level shifting, amplification and noise reduction
 - Generation of a suitable interrupt to trigger measurement from the microprocessor.
- Practical operation from a single 5V power regulator.



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Practical Signal Conditioning Circuits

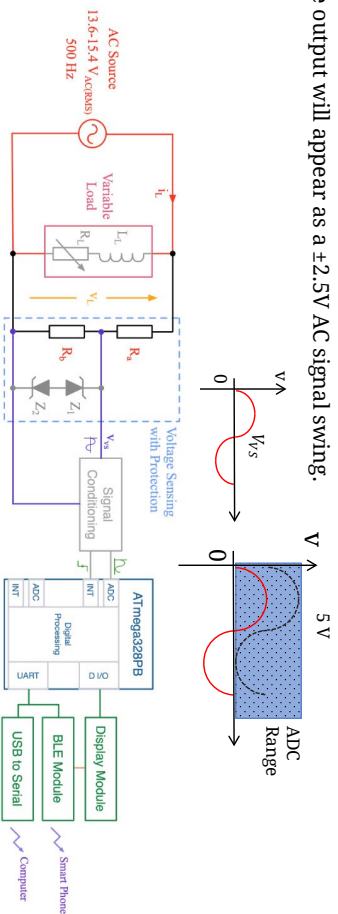
Level Shifting
Thevenin Equivalents
Operation with Single Sided supplies

Signal Conditioning of the Voltage and Currents

The Analogue to Digital Converter (ADC) measurement in a microprocessor only samples positive voltages (0-5V), while the AC signal has no DC component. We also normally only have one single supply to operate the op-amp off.

Therefore we need to

- Add a DC offset to the signal, then amplify and filter as required
- Use an op-amp that is operating off a single 5V supply.
- The output will appear as a $\pm 2.5V$ AC signal swing.



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Level Shifting and Amplifying using an Op-Amp with a Single Supply

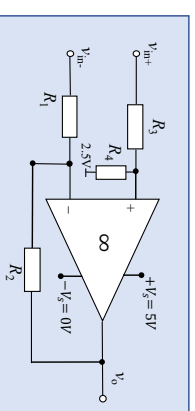
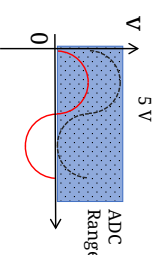
Using superposition $v^{(+)}$ can be found in terms of v_{in+} and $2.5V$ as:

$$v^{(+)} = \frac{R_4}{(R_3+R_4)} v_{in+} + \frac{R_3}{(R_3+R_4)} 2.5 = \frac{1}{1+\frac{R_4}{R_3}} \left(\frac{R_4}{R_3} v_{in+} + 2.5 \right)$$

Show that the difference amplifier expressions are now:

$$v_o = \left(1 + \frac{R_2}{R_1} \right) \left(\frac{1}{1+\frac{R_4}{R_3}} \right) \left(\frac{R_4}{R_3} v_{in+} + 2.5 \right) - \frac{R_2}{R_1} v_{in-}$$

If $\frac{R_4}{R_3} = \frac{R_2}{R_1}$ then: $v_o = A_{id} v_{id} + 2.5$ where $A_{id} = \frac{R_2}{R_1}$



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The Thevenin Equivalent of the Voltage

- The voltage v_{vs} effective series impedance must be considered in the signal conditioning. This can be assessed using the Thevenin equivalent source.

- As discussed earlier

$$v_{vs} = v_L \frac{R_b}{(R_a + R_b)}$$

This has a Thevenin equivalent series source impedance based on short circuiting v_L

Given by:

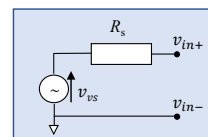
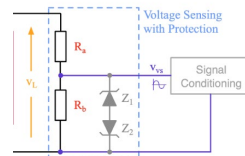
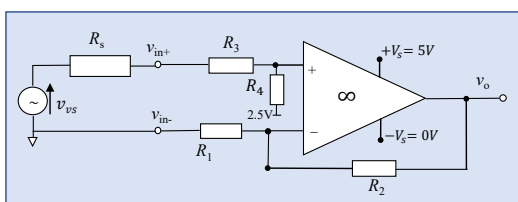
$$R_s = R_a \parallel R_b$$

R_s must be included in the difference amplifier design since now $R'_3 = R_s + R_3$

Select:

$$R'_3 = R_1$$

$$R_4 = R_2$$



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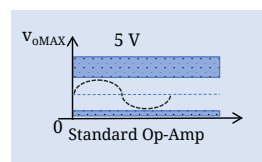
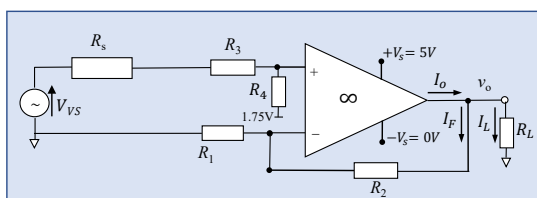
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Output Voltage limitations of Op-Amps

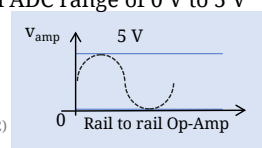
Standard Op-Amps have outputs that cannot reach voltage supply rails, and current limits when sinking or sourcing.

Voltage limits: The upper and lower voltage limits are L_+ and L_- respectively. Trying to output a voltage above or below these limits will cause the signal to be clipped.

- E.G., the usable output range of the LM324/358 (in your design) has $L_- = 5mV$, $L_+ = 3.5V$ (when the supply is 5 V)



- A rail to rail Op-Amp could be used to improve accuracy
 - Output could reach close to supply rails thus allowing to utilize full ADC range of 0 V to 5 V



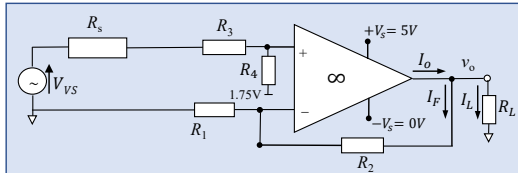
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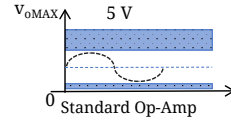
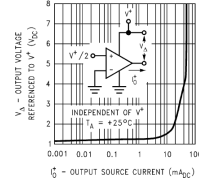
Output Current limitations of Op-Amps

Current Limits: the specified current limits (typically 20mA), include the total of the load and feedback circuit which the op-amp must supply, but the capability of the 358 to sink current at low voltage is poor.

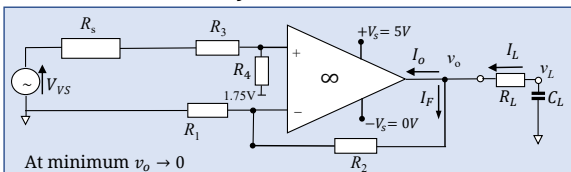
- If there is an output load R_L , when does the maximum current flow?
 - At the peak voltage (3.5V), $I_{oMAX} = I_F + I_L$. If $I_{oMAX} > I_{LIM}$ then the waveform will clip. Choose resistors in $k\Omega$



Output Characteristics Current Sourcing

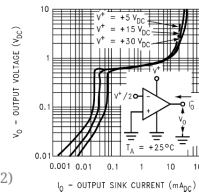


- If there is an output filter there may be issues sinking the current into the op-amp to bring the load voltage down to $< 1V$ that may mean we need smaller values for R_L , R_2 and R_1 to reduce the capacitor voltage in time



At minimum $v_o \rightarrow 0$

Output Characteristics Current Sinking



In general, choose:
 $R_s < 500k$
 $C_s > 100pF$

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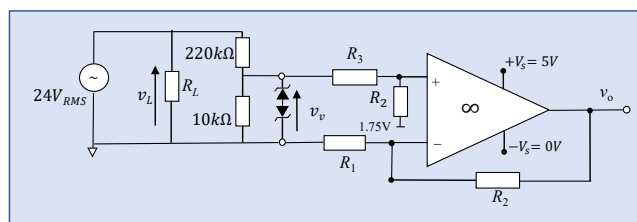
Ex: Voltage Measurement

Design a differential amplifier, that can level shift v_v measured by the voltage sensing circuit in the figure below to be within 0 to 3.5 V (within the op-amp and ADC range). Ignore the lower limitation on the Op-Amp.

- Design for a level shift of $3.5V/2 = 1.75V$

- Peak to peak level of sensed signal

$$v_{v(pk-pk)} = 24 \times 2 \times \sqrt{2} \times \frac{10k}{220k+10k} = 2.95V$$



The differential amplifier gain should be no more than 1.4, but choosing $A_{id} \sim 1$ is sufficient

- Aim to achieve a 3 V signal swing at output, (0.25 V off the rail limit).
- The RMS Thevenin voltage source and resistance are: $v_{vs} = 24 \times \frac{10k}{220k+10k} = 1.04V_{RMS}$ and $R_s = \frac{220k \times 10k}{220k+10k} = 9.6k\Omega$
- If $A_{id} = \frac{R_2}{R_1} = 1$, then make $R_1 = R_2 = (R_s + R_3)$.
- We can choose $R_1 = R_2 = 100k\Omega$ and $R_3 = 90.4k\Omega \approx (22k\Omega + 68k\Omega)$ by connecting two E12 value resistors in series.

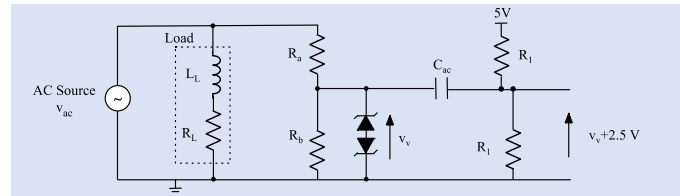
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Level Shifting with a Cap

A DC bias derived through for example a voltage bias and an AC coupling capacitor can be used for level-shifting

- An example of voltage measurement is shown in figure
 - An AC coupling capacitor, C_{ac} , used to add the sensed voltage, v_v , with a 2.5V bias derived through two R_1 's



C_{ac} together with Thevenin resistances of v_v and bias voltage all together form a high-pass filter

This arrangement can be used to eliminate the need for an Op-Amp

- The disadvantage is that this network cannot amplify the measured signal and will couple any high-frequency noise present, which is automatically suppressed by the op-amp due to its internal filtering.
- The capacitor can introduce some phase shift in the signal.

Ex: Current Measurement

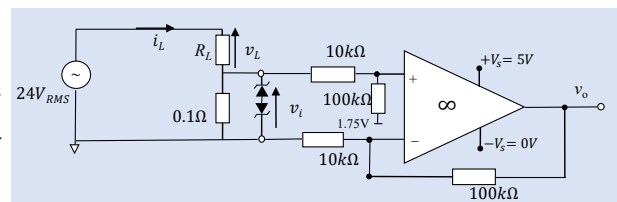
A 100 mΩ shunt resistor is used to sense the current flowing through a variable load. The load current is in the range 0.2 A_{RMS} to 1 A_{RMS}.

Design a differential amplifier, that can amplify and level shift this signal to be within 0 to 3.5 V (op-amp range).

- Max and min peak-peak signal sensed by the shunt

$$v_{i,MAX(pk-pk)} = i_{(pk-pk)} R_s = 2\sqrt{2} \times 1 A_{RMS} \times 100m\Omega = 282mV$$

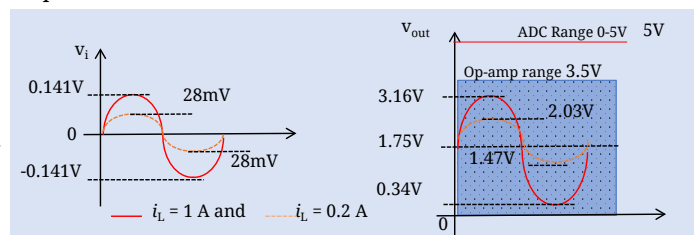
$$v_{i,MIN(pk-pk)} = i_{(pk-pk)} R_s = 2\sqrt{2} \times 0.2 A_{RMS} \times 100m\Omega = 56mV$$



Again design to achieve around a 3 V signal swing at output (0.25 V off either rail)

$$A_{id} = \frac{R_2}{R_1} = \frac{3V}{0.282V} = 10.64$$

- Choosing a gain of $A_{id} \sim 10$ would satisfy this.
- Choose $R_1 = 10k\Omega$ and $R_2 = 100k\Omega$ (E12 series)
- Min and Max voltage swings are: 0.56V and 2.82V



RC Filters and the Frequency Response

A simple RC Circuit

Frequency domain Bode plots

Frequency domain filtering

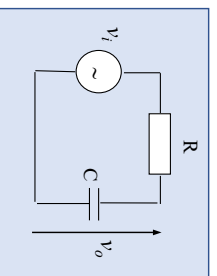
A Simple RC Circuit



A simple RC circuit is shown below.

- The gain of this circuit is complex and dependent on frequency because of the capacitor
- Recall from Electeng291 that using *Laplace*, the impedance of a capacitor is given by $Z_c = \frac{1}{j\omega C}$
- To find the steady state gain of the circuit in the frequency domain - analyse as a voltage divider:

$$\frac{V_o(\omega)}{V_i(\omega)} = \frac{Z_c}{Z_R + Z_c} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega CR}$$



Frequency Domain Bode Plots

If the gain of a circuit is complex, then like any complex number it can be expressed in terms of its magnitude and phase.

- For $z = a + jb$

The magnitude of this complex number is: $|z| = |a + jb| = \sqrt{a^2 + b^2}$

The phase of this complex number is: $\angle z = \tan^{-1}\left(\frac{b}{a}\right)$

- In the case of the complex gain expression for the RC circuit, both phase and magnitude are changing with ω

Bode plots are used to view separately how the *magnitude* and *phase* vary with frequency.

However it is convenient to look at the impact using a logarithmic scale because the impact can be seen over a very wide frequency range - often over many orders of magnitudes. Our ear also has a logarithmic response which makes this assessment useful. The “bel” scale was defined as the log-base-ten of the ratio of two signal intensities related to the power or energy in the signal. As there are 10 decibels per bel the power ratio is defined as:

$$10 \log_{10} \frac{P_o}{P_i} \qquad \text{the power ratio in decibels (dB)}$$

$$20 \log_{10} \frac{V_o}{V_i} \quad (\text{the voltage ratio in dB})$$

Power is proportional to the square of the voltage and therefore this is re-expressed as

Here the voltages are in terms of their *rms* quantities.

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Frequency Domain Bode Plots

In engineering base 10 is assumed as the default when $\log H(\omega)$ is used.

- The Bode magnitude is a plot of the gain expression in dB $20 \log \left| \frac{V_o(\omega)}{V_i(\omega)} \right|$ against $\log \omega$

- The Bode phase is a plot of $\tan^{-1}\left(\frac{V_o(\omega)}{V_i(\omega)}\right)$ against $\log \omega$

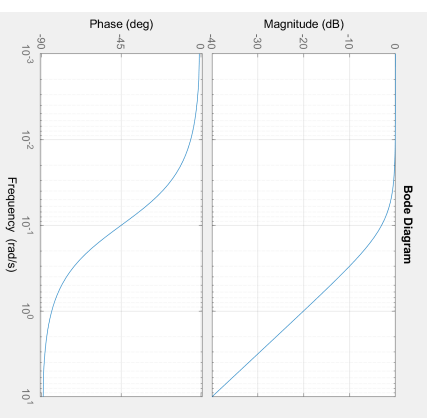
Simple log rules are useful to remember

- $\log AB = \log A + \log B$ and $\log A/B = \log A - \log B$

Creating the **Bode** plot

- Defining the time constant $T = RC$, then $G(\omega) = \frac{1}{1+j\omega T}$
- *Matlab* can be used to plot the Bode expression for any given value of T .

```
clc
T=10; % Time constant = 10s
num=1; % define numerator as 1
den=[T, 1]; % define denominator as sT+1 = j\omega T+1
sys = tf(num,den) % define the system transfer function
figure(1) % set up a figure to plot the system
bode(sys) % create the bode plot
grid % add a grid
```



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Frequency Domain Bode Plots

Sketching approximate Bode plots

For $G(\omega) = \frac{1}{1+j\omega T}$,

Bode magnitude in dB is: $|G(\omega)|_{dB} = 20 \log \left| \frac{1}{1+j\omega T} \right| = -20 \log |1 + j\omega T| = -20 \log |1 + j\omega/(1/T)|$

Bode phase is: $\angle G(\omega) = \angle \frac{1}{1+j\omega T} = -\angle(1 + j\omega T) = -\tan^{-1}(\omega T/1) = -\tan^{-1}(\omega/(1/T))$

The breakpoint frequency can be defined as $\omega_b = 1/T$

- When $\omega = \omega_b$ then $|G(\omega_b)|_{dB} \cong -20 \log |1 + j1| = -20 \log |\sqrt{2}| = -3dB$ $\angle G(\omega) = -\tan^{-1}(1/1) = -45^\circ$

Asymptotes can be determined either side of ω_b

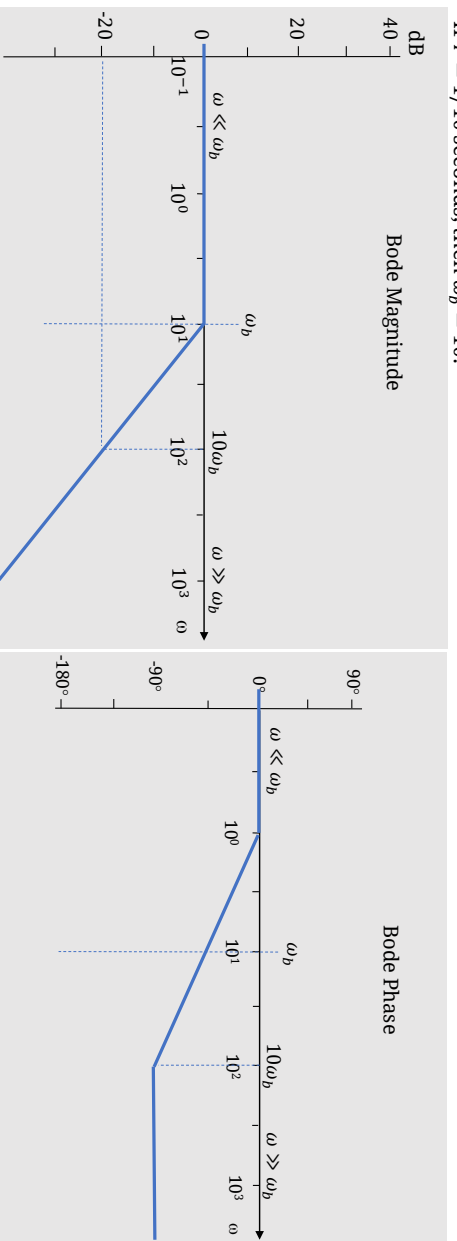
- When $\omega \ll \omega_b$, then $|G(\omega)|_{dB} = -20 \log |1 + j\omega/\omega_b| \cong -20 \log |1| = 0dB$ $\angle G(\omega \ll \omega_b) = -\tan^{-1}(0/1) = 0$
 - When $\omega \gg \omega_b$, then $|G(\omega)|_{dB} = -20 \log |1 + j\omega/\omega_b| \cong -20 \log |j\omega/\omega_b|_{dB}$ $\angle G(\omega \gg \omega_b) = -\tan^{-1}(\omega/\omega_b) = -90^\circ$
- Example: $\omega = 10\omega_b$ gives $|G(10\omega_b)|_{dB} = -20 \log |10\omega_b/\omega_b| = -20dB$, $\angle G(10\omega_b) = -\tan^{-1}(10) = -84.3^\circ$
- $\omega = 100\omega_b$ gives $|G(100\omega_b)|_{dB} = -20 \log |100| = -40dB$, $\angle G(100\omega_b) = -\tan^{-1}(100) = -89.4^\circ$

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Frequency Domain Bode Plots

If $T = 1/10$ seconds, then $\omega_b = 10$.



Maximum magnitude error is 3dB at the break point frequency, while in the *phase* plot, the max error from a straight line approximation = 5.7°

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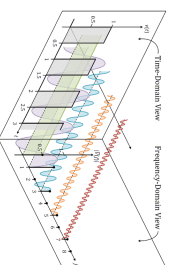
Frequency Domain Filtering

All real world signals are made up of various frequencies.

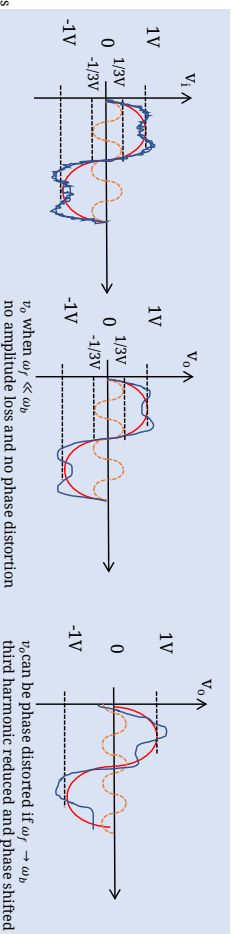
From Electeng291 notes recall that a periodic waveform in the time domain is constructed of various frequency components – and the magnitude of these can be determined from the Fourier series:

For example consider the square wave shown. When this waveform is passed through the filter:

- Frequencies at least a decade below $\omega_b = 2\pi f_b$ will have their magnitude and phase preserved.
- Frequencies above ω_b will have their magnitude reduced (attenuated) and will also be shifted in phase.
- Ex: Assume an input waveform as: $v_i = \sin \omega_f t + 0.33 \sin 3\omega_f t + \text{noise}$



Ex: Square wave time domain v.s. frequency component amplitudes & phases



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Questions?